Model performance is evaluated using the Mean Squared Error (MSE), which measures the average squared difference between actual and predicted values: # Evaluation

```
mse = mean_squared_error(y_test, predictions)
print("Mean Squared Error:", mse)
```

Example Output

For the dataset provided, the output might look like this: Mean Squared Error: 1.2e+09

This indicates the average squared error in the predictions. A lower MSE signifies better performance.

Insights

Linear regression is straightforward and interpretable, making it a great starting point for regression problems. Key insights include:

- **Interpretability:** The model coefficients indicate how much the target variable changes with a unit change in the input features.
- **Limitations:** Linear regression assumes a linear relationship between variables. If the relationship is non-linear, alternative models such as polynomial regression or tree-based methods may be more appropriate.
- **Scalability:** It performs well on datasets with a large number of features but requires feature scaling if the features have different ranges.

When to Use Linear Regression

Linear regression is best suited for problems where:

- There is a clear linear relationship between the target variable and input features.
- The dataset is free of multicollinearity (high correlation between independent variables).